Codility

Placement Calendar - 2

app.codility.com/c/run/FMQADK-DA4/

Oct 4 10:07 AM

1 Task 1

2

1h 34min

C++

Examples:

1. Given A = [1, 1, 3], B = [2, 2, 1] and S = 3, the function should return `true`. We could assign patients in the following way: [1, 2, 3], where the K-th element of the array represents the number of the slot to which patient K was assigned. Another correct assignment would be [2, 1, 3]. On the other hand, [2, 2, 1] would be an incorrect assignment as two patients would be assigned to slot 2.

2. Given A = [3, 2, 3, 1], B = [1, 3, 1, 2] and S = 3, the function should return `false`. There are only three slots available, but there are four patients who want to visit the doctor. It is therefore not possible to assign the patients to the slots so that only one patient at a time would visit the doctor.

3. Given A = [2, 5, 6, 5], B = [5, 4, 2, 2] and S = 8, the function should return `true`. For example, we could assign patients in the following way: [5, 4, 6, 2].

4. Given A = [1, 2, 1, 6, 8, 7, 8], B = [2, 3, 4, 7, 7, 8, 7] and S = 10, the function should return `false`. It is not possible to assign all of the patients to one of their preferred slots so that only one patient will visit the doctor during one slot.

Write an **efficient** algorithm for the following assumptions:

Files

task1

solution.cpp

test-input.txt

solution.cpp

Submit Task

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

bool solution(vector<int> &A, vector<int> &B, int S) {
    // write your code in C++14 (g++ 6.2.0)
}

```

Test Output

Run Tests

Any problems with the editor? Switch to basic editor

Give Feedback

Autocomplete is connected | All changes saved

DELL

Codility

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Placement Calendar - 2

Oct 4 10:07 AM

1 Task 1

2

C++

1h 34min

to slot 2.

2. Given A = [3, 2, 3, 1], B = [1, 3, 1, 2] and S = 3, the function should return `false`. There are only three slots available, but there are four patients who want to visit the doctor. It is therefore not possible to assign the patients to the slots so that only one patient at a time would visit the doctor.

3. Given A = [2, 5, 6, 5], B = [5, 4, 2, 2] and S = 8, the function should return `true`. For example, we could assign patients in the following way: [5, 4, 6, 2].

4. Given A = [1, 2, 1, 6, 8, 7, 8], B = [2, 3, 4, 7, 7, 8, 7] and S = 10, the function should return `false`. It is not possible to assign all of the patients to one of their preferred slots so that only one patient will visit the doctor during one slot.

Write an **efficient** algorithm for the following assumptions:

- N is an integer within the range [1..100,000];
- S is an integer within the range [2..100,000];
- each element of arrays A, B is an integer within the range [1..S];
- no patient has two preferrences for the same slot, i.e. A[i] ≠ B[i].

Files

task1

solution.cpp

test-input.txt

solution.cpp ×

Submit Task

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

bool solution(vector<int> &A, vector<int> &B, int S) {
    // write your code in C++14 (g++ 6.2.0)
}
```

Test Output

Run Tests

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Codility

Placement Calendar - 2

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Oct 4 10:07 AM

1

2

Task 2

C++

1h 34min

Once constructed, a turbine occupies exactly one cell of the map and each cell can contain at most one turbine. The distance between two locations is measured as the sum of horizontal and vertical distances between two cells on the map. For example, the distance between cells A[3][2] and A[1][4] is |1 - 3| + |4 - 2| = 4.

Write a function:

```
int solution(vector< vector<int> > &A, int K);
```

that, given matrix A consisting of N rows and M columns and an integer K, returns the maximum number of wind turbines that can be built while following the environmentalists' advice.

Examples:

1. Given A of size 4 × 3 and K = 1 with bird habitats at (0, 0), (2, 2) and (3, 1):

```
A[0][0] = 1   A[0][1] = 0   A[0][2] = 0
A[1][0] = 0   A[1][1] = 0   A[1][2] = 0
A[2][0] = 0   A[2][1] = 0   A[2][2] = 1
A[3][0] = 0   A[3][1] = 1   A[3][2] = 0
```

the function should return 3. Wind turbines can be built in three locations: (0, 2), (1, 1) and (2, 0).

Files

task2

solution.cpp

test-input.txt

solution.cpp

Submit Task

```
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}
```

Test Output

Run Tests

Autocomplete is connected | All changes saved

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Codility

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Task 2

C++

1h 34min

The effects of global warming are becoming more and more noticeable each year. That is why many countries aim to reduce greenhouse gas emissions by generating electricity from the wind instead of from fossil fuels. Wind turbines have fewer negative effects on the environment and emit less carbon dioxide.

A company has got a contract to build a wind farm on an enclosed, rectangular area of land. The goal is to build as many turbines as possible, to maximize the wind farm's power capacity.

It was recently discovered that the land is inhabited by an endangered species of bird. Thus, the company wants to build the wind turbines at a distance greater than K from every habitat.

The map of the land is given as a rectangular matrix A of integers, where birds' habitats are represented by 1 and all other locations are represented by 0.

Once constructed, a turbine occupies exactly one cell of the map and each cell can contain at most one turbine. The distance between two locations is measured as the sum of horizontal and vertical distances between two cells on the map. For example, the distance between cells A[3][2] and A[1][4] is |1

Files

task2

solution.cpp

test-input.txt

solution.cpp

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}
```

Submit Task

Test Output

Run Tests

Autocomplete is connected | All changes saved

Codility

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Placement Calendar - 2

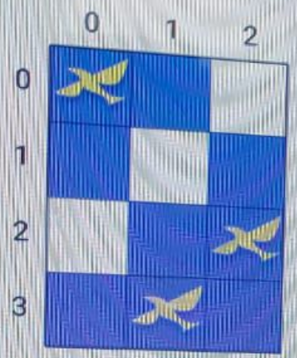
Oct 4 10:07 AM

Task 2

C++

1h 34min

the function should return 3. Wind turbines can be built in three locations: (0, 2), (1, 1) and (2, 0).



2. Given A of size 3 × 4 and K = 2 with no habitats:

```
A[0][0] = 0    A[0][1] = 0    A[0][2] = 0    A[0][3] = 0
A[1][0] = 0    A[1][1] = 0    A[1][2] = 0    A[1][3] = 0
A[2][0] = 0    A[2][1] = 0    A[2][2] = 0    A[2][3] = 0
```

the function should return 12. A wind turbine can be built in every location.

0 1 2 3

Files

task2

solution.cpp

test-input.txt

solution.cpp ×

✓ Submit Task

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}
```

Test Output

Run Tests

Any problems with the editor? Switch to basic editor

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Task 1

C++

There are N patients (numbered from 0 to N-1) who want to visit the doctor. The doctor has S possible appointment slots, numbered from 1 to S. Each of the patients has two preferences. Patient K would like to visit the doctor during either slot A[K] or slot B[K]. The doctor can treat only one patient during each slot.

Is it possible to assign every patient to one of their preferred slots so that there will be at most one patient assigned to each slot?

Write a function:

```
bool solution(vector<int> &A, vector<int> &B, int S);
```

that, given two arrays A and B, both of N integers, and an integer S, returns `true` if it is possible to assign every patient to one of their preferred slots, one patient to one slot, and `false` otherwise.

Examples:

1. Given A = [1, 1, 3], B = [2, 2, 1] and S = 3, the function should return `true`. We could assign patients in the following way: [1, 2, 3], where the K-th element of the array represents the number of the slot to which patient K was assigned. Another correct

Files

task1

solution.cpp

test-input.txt

solution.cpp

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

bool solution(vector<int> &A, vector<int> &B, int S) {
    // write your code in C++14 (g++ 6.2.0)
}
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Test Output

Run Tests

Submit Task

1h 34min

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Placement Calendar - 2

Task 2

C++

1h 33min

the function should return 40. Wind turbines can be built in the white cells. Cells that are too close to birds' habitats are marked in blue.

Write an **efficient** algorithm for the following assumptions:

- N and M are integers within the range [1..600];
- K is an integer within the range [1..1,200];
- each element of matrix A is an integer that can have one of the following values: 0, 1.

Files

task2

solution.cpp

test-input.txt

solution.cpp

Submit Task

```cpp
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}

```

Test Output

Run Tests

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Placement Calendar - 2

Oct 4 10:08 AM

Task 2

1h 33min

C++

Files

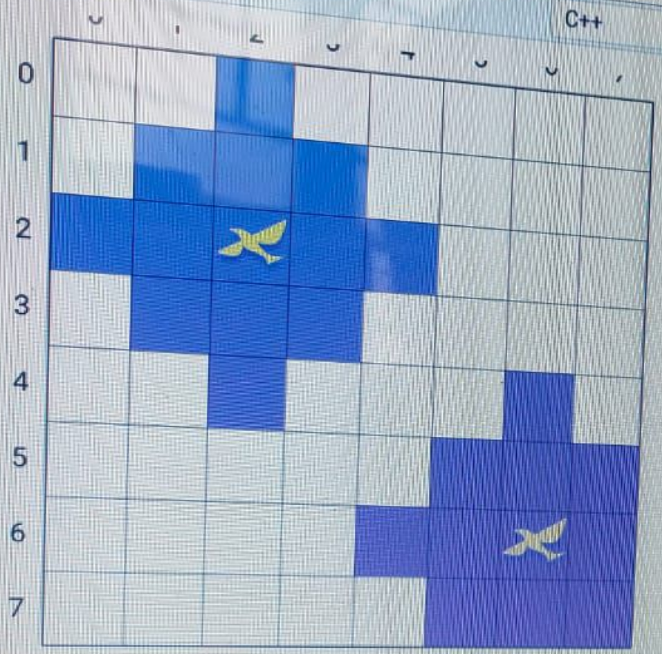
task2

solution.cpp

test-input.txt

solution.cpp

Submit Task



```
// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}
```

the function should return 40. Wind turbines can be built in the white cells. Cells that are too close to birds' habitats are marked in blue.

Write an **efficient** algorithm for the following assumptions:

- N and M are integers within the range [1..600];

Test Output

Run Tests

Any problems with the editor? Switch to basic editor

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Codility

Placement Calendar - 2

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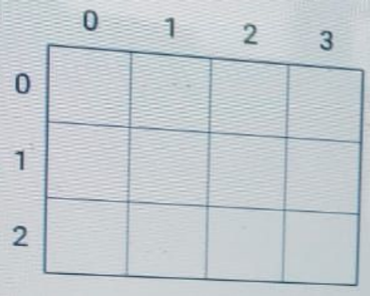
Task 2

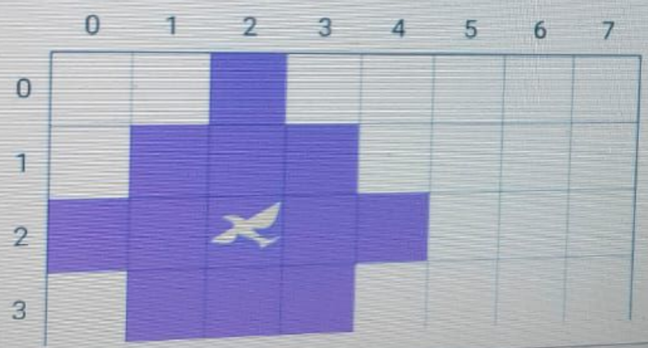
C++

1h 33min

the function should return 12. A wind turbine can be built in every location.



3. Given A of size 8 × 8 and K = 2 with bird habitats at (2, 2) and (6, 6):



Files

task2

solution.cpp

test-input.txt

solution.cpp

Submit Task

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// you can use includes, for example:
// #include <algorithm>

// you can write to stdout for debugging purposes, e.g.
// cout << "this is a debug message" << endl;

int solution(vector< vector<int> > &A, int K) {
    // write your code in C++14 (g++ 6.2.0)
}

```

Test Output

Run Tests

Autocomplete is connected | All changes saved

DELL